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## Unit – 1 : Session – 2 : SLO – 2 | BCD, Gray code, Excess-3

### ■ Multiple Choice Questions

1. c — 0110 (add 6 correction)
2. c — Copy the first bit
3. c — 1011 (decimal 11 is not a valid BCD digit)
4. c — Better for analog-to-digital conversion (only one bit changes between consecutive values)
5. b — Handling sums greater than 9 in BCD

### ■ Two-Mark Questions

1. BCD (Binary-Coded Decimal) represents each decimal digit 0–9 with its own 4-bit binary code (8421 weighted). Application: digital clocks/calculators and financial systems where exact decimal display is needed.
2. Gray to binary (5-bit): (i) Copy the MSB of Gray as MSB of binary. (ii) Each next binary bit = previous BINARY bit XOR current Gray bit. (iii) Repeat left to right for all 5 bits.

### ■ Scenario-Based Long Answer

- Correction rule: after adding two BCD digits, if the 4-bit result is greater than 1001 (9) or a carry is generated, add 0110 (6) to that digit and propagate the carry to the next higher digit.
- Example:  $9 + 3$ :  $1001 + 0011 = 1100$  (12, invalid BCD). Add 0110  $\rightarrow 1100 + 0110 = 1\ 0010$ . Result: 0010 in the units digit and a carry of 1 to the tens digit, i.e.  $0001\ 0010 = 12$  in BCD. The clock's display logic applies this per digit so every digit stays a valid 0–9 code.

### ■ Match the Following

- BCD — a) 8,4,2,1 weighted code
- Excess-3 — b) Non-weighted, self-complementary code
- Gray Code — c) Used for minimizing errors in A/D conversion
- Invalid BCD Code — d) 1011
- Binary to Gray Conversion — e) Copy first bit, then XOR

### ■ Fill in the Blanks

1. 0111 2. 0110 3. MSB (first/most significant) 4. 0110 5. errors

### ■ Scenario-Based Small Activity

- Convert binary 1010 to Gray: (i) Copy first bit:  $g_3 = 1$ . (ii)  $g_2 = b_3 \text{ XOR } b_2 = 1 \text{ XOR } 0 = 1$ . (iii)  $g_1 = b_2 \text{ XOR } b_1 = 0 \text{ XOR } 1 = 1$ . (iv)  $g_0 = b_1 \text{ XOR } b_0 = 1 \text{ XOR } 0 = 1$ .
- Output Gray code = 1111. Because only one bit changes between consecutive Gray values, transmitting 1111 avoids multi-bit transition errors from the sensor.

### ■ True or False

1. FALSE — Only one bit changes between consecutive Gray values.
2. FALSE — BCD is not efficient for high-speed arithmetic (needs corrections).
3. TRUE
4. FALSE — Bits are XORed, not reversed.
5. TRUE